

Ideation Phase

Brainstrom & Idea Prioritization

Date	October 03,2026
Team ID	SWTID-2026-7210
Project Name	Script Controlled ACL: Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions to secure data object visibility and control access metrics dynamically.

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Problem Statement Definition (How Might We):

How might we design a robust, granular, and scalable Script-Controlled Access Control List (ACL) system that dynamically restricts or grants record-level access based on real-time data field values, ensuring complete data isolation and privacy compliance across multi-tenant corporate platforms?

Step-2: Brainstorm, Idea Listing and Grouping

Idea Category / Cluster	Brainstormed Solutions (Sticky Notes Equivalent)	Implementation Strategy & Mechanism
Category A: Server-Side Scripted Triggers	Idea 1: Implement dynamic ServiceNow-style Advanced Scripted ACLs using glidesystem checks. Idea 2: Utilize conditional Database View constraints evaluating operational column data values natively.	Executes directly before rendering data records to users. Evaluates JavaScript rules synchronously inside the secure platform wrapper.
Category B: API Gateway & Policy Filters	Idea 3: Build an Attribute-Based Access Control (ABAC) JSON-policy validator inside an API middleware layer. Idea 4: Inject row-level contextual request headers dynamically based on Active Directory field profiles.	Extracts user claims and evaluates field attributes externally at the routing entry point, stripping unauthorized records before client delivery.

Category C: Encryption & Masking Controls	Idea 5: Leverage field-value driven Cryptographic Key Derivation where row keys change relative to field status. Idea 6: Deploy dynamic data masking routines on confidential columns when state constraints fail.	Blurs the data payload automatically if field conditions do not perfectly align with the actor's clearance parameters.
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Step-3: Idea Prioritization (Feasibility vs Importance Framework)

To maximize short-term impact while ensuring structural longevity, the team placed brainstormed options across a four-quadrant prioritization matrix tracking organizational Business Importance against Technical Feasibility.

Prioritization Matrix Mapping Evaluation:

Proposed Solution Component	Business Importance	Technical Feasibility	Strategic Priority Action
Advanced Scripted Server-Side ACLs	HIGH (Ensures rigorous security compliance at object layer)	HIGH (Native engine support, minimal code layout footprint)	Immediate Implementation (Quick Win / High Value)
Attribute-Based API Policy Gateway Middleware	HIGH (Protects downstream cloud endpoints decoupled from system core)	MEDIUM (Requires network infrastructure configuration and API latency tuning)	Phase 2 Target (Plan & Resource)
Field-Value Cryptographic Key Derivation	MEDIUM (Provides extreme defense-in-depth security)	LOW (High processing overhead and key management complexity)	Postpone / Defer (Low ROI for current prototype phase)

Conclusion & Selected Core Project Path

Selected Idea: The team has unanimously selected Script-Controlled Server-Side ACLs as the primary baseline solution. This methodology yields the optimal overlap of maximum data confidentiality (High Importance) and rapid native implementation flow (High Feasibility). The next phase will progress into detailed architectural component mapping, sequence diagram development, and database schema prototyping.